MADHURI MERUGU

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Professional Summary

Computer Science graduate with hands-on software engineering experience at Swiggy as an Associate Software Development Engineer. Worked on Java-based backend development, service flow improvements, error handling, and catalog-related functionality. Familiar with Java, SQL, Spring Boot, Object-Oriented Programming, and Data Structures. Interested in backend development and building reliable software applications.

Education

G. Narayanamma Institute of Technology and Science — 2023 – 2026 | CGPA: 9.47
Bachelor of Technology, Computer Science and Engineering — Hyderabad, Telangana, India

Government Polytechnic for Women — 2020 – 2023 | CGPA: 9.8
Diploma in Computer Engineering — Jogipet, Sangareddy, Telangana

Telangana State Model School — 2019 – 2020 | CGPA: 9.8
Secondary Education (Class X) — Chityal, Telangana, India

Experience

Swiggy - Associate Software Development Engineer

Jan 2026 – October 2026

- Worked on Java-based backend development for Instamart discovery-related services and functionality.
- Contributed to Taxonomy Flow migration, helping reduce latency and minimize dependencies on multiple internal services.
- Implemented error handling and user-friendly error messaging across Category Listing, Campaign Listing, MxN, and Collections flows.
- Worked on out-of-stock handling for collections by filtering unavailable items and excluding widgets that did not meet the minimum item threshold

Skills

Technical Skills: Java, SQL, Data Structures & Algorithms, Object-Oriented Programming, DBMS, Git, GitHub, Basics of Artificial Intelligence and Machine Learning

Professional Skills: Problem Solving, Analytical Thinking, Team Collaboration, Communication, Time Management, Quick Learning

Projects

1. Smart Helmet for Coal Miners

Technologies: *IoT, Arduino Uno, Embedded Systems, Sensors, RF Communication*

- Developed an IoT-based smart helmet prototype to monitor environmental conditions and improve miner safety.
- Integrated sensors to detect hazardous gases and environmental changes, with alert mechanisms using RF communication.
- Designed real-time monitoring and warning functionality to notify users about potentially unsafe conditions.

2.Ground Water Level Prediction using Machine Learning

Technologies: *Python, Flask, Scikit-learn, Pandas, NumPy, Matplotlib*

- Developed a machine-learning application to predict groundwater levels using historical groundwater and climate-related data.
- Performed data preprocessing, feature engineering, and visualization to prepare the dataset for model training.
- Implemented and evaluated Logistic Regression, Decision Tree, and Random Forest models to classify groundwater levels into Low, Normal, and High categories.
- Developed a Flask-based interface to display prediction results.

Certificates

- Indo-Dutch Cyber Security School, 2024
- AI & ML for Geo Data Analysis –ISRO, 2024
- Cloud Computing Fundamentals - Google Cloud, 2024
- Cyber Security Essentials – CISCO, 2025

Achievements

- 1st Place - Poster Presentation, **NEXORA National-level Technical Fest**, Jan 2025
  Demonstrated innovation and analytical depth in presenting a technical solution, outperforming peer participants at a competitive national forum.
- Telangana State Rank - **936, TS-ECET** 2023
  Secured a top 1000 rank in the Telangana State Engineering Common Entrance Test .